BOARD SIZE (STANDARD 60 + 120 + 60 CM)

PROJECT BOARD - STANDARD SIZE

Yaai's Journal: A Southern-Thailand Adventure

BOARD 1	BOARD 2	BOARD 3
60 cm	120 cm	60 cm
Background, Objectives, Hypothesis	Process, Results, Discussion	Performance, Utilization, Recommendation

60 cm | 120 cm | 60 cm

Total 240 cm wide, 120 cm high - three panels, English throughout.

THEME: WHITE AND RED

THEME: WHITE AND RED

Yaai's Journal: A Southern-Thailand Adventure

A learning game for Python and the Southern coast

White | Red | Midnight blue | Lantern gold

Headline TH SarabunPSK 18 pt bold - body TH SarabunPSK 16 pt

EXAMPLE: THE FINISHED BOARD

EXAMPLE: THE FINISHED BOARD - THREE PANELS, 240 x 120 cm



BOARD 1



BOARD 2



BOARD 3

Rendered from the delivered board file (PDF export of the same document).

Board 1

ai's Journal: A Southern-Thailand Adventure that Teaches Python"

BACKGROUND AND SIGNIFICANCE

Our syllabus names Python as the first real language, but time to write programs is short and the exercises arrive with no reason to care about the answer. Our own region reaches us the same way: the coast from Chumphon to Narathiwat as a short passage that gives no picture.

Yaai's Journal was built as the answer: a two-dimensional adventure in which every reward is earned with a program. The player travels with the puppet teacher Noi to restore the 24 pages of Yaai's journal, torn away by a storm; at each stop a rune tablet asks for a small, real Python program, checked automatically, before the next page can be read. Five acts now run from the Gulf to the deep South, with 40 landmarks, 25 side quests and 74 encounters. It installs from two pip packages and a 201-case suite guards it.

OBJECTIVES

This project aims to:

1.To develop Yaai's Journal, a 2-D adventure game that teaches Python step by step, page by page, through its story.

2. To teach 24 Southern Thai words, 40 landmarks and the history of the coast inside the same quests that teach code.

HYPOTHESIS

Because the starter code on every tablet is written to fail its own validator, the story cannot advance until the player writes correct Python. The 24 solutions we prepared pass the same validator, and the 201-case automated test suite keeps the program working after every edit.

Board 2

PROCESS

Built in five phases, each ending with something the test suite could check.

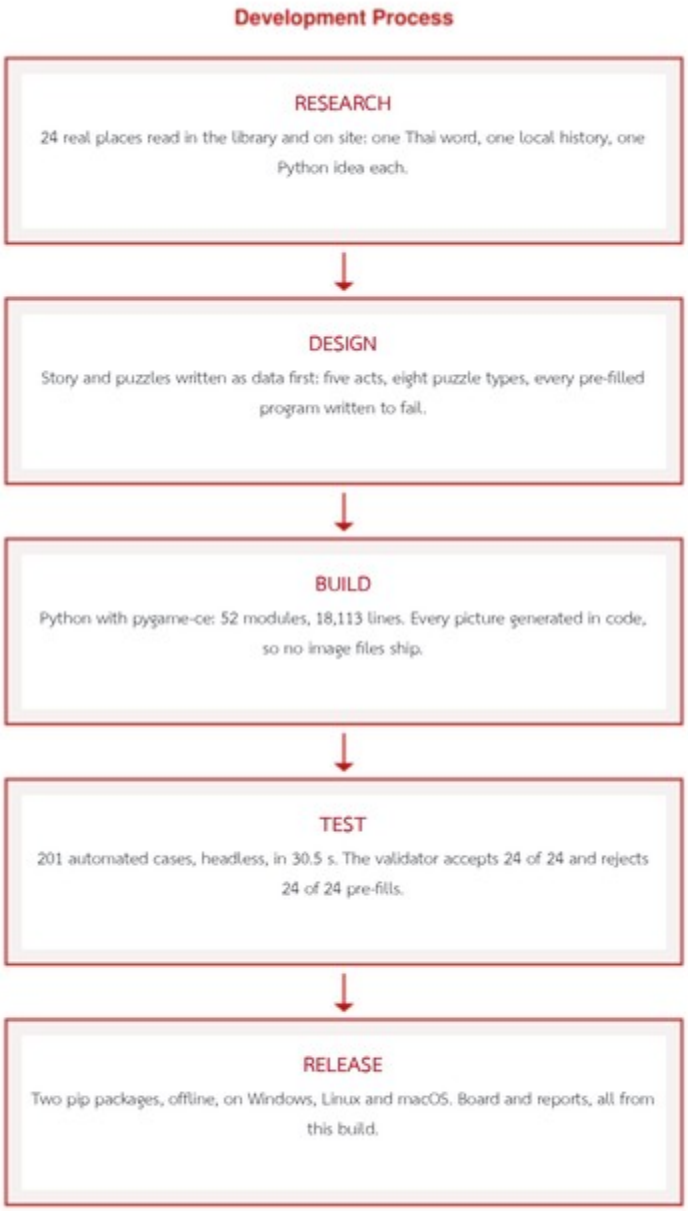
1. Research - 24 real places read up in the library and on site: each stop keeps one Thai word, one piece of local history and one Python idea from our syllabus.

2. Design - story and puzzles written as data first: five acts, eight puzzle archetypes, and the rule that every pre-filled program must fail its own validator.

3. Build - Python with pygame-ce and Pillow: 52 modules and 18,113 lines, with every picture generated in code, so no image files ship.

4. Test - 201 automated cases in 26 test files run headless in 30.5 s; the validator accepts all 24 solutions and rejects all 24 pre-filled starters.

5. Release - two pip packages, offline, on Windows, Linux and macOS; this board, the 29-page report and the 25-page Thai report all come from this build.



Picture 1 Yaai's Journal development process (research to release)

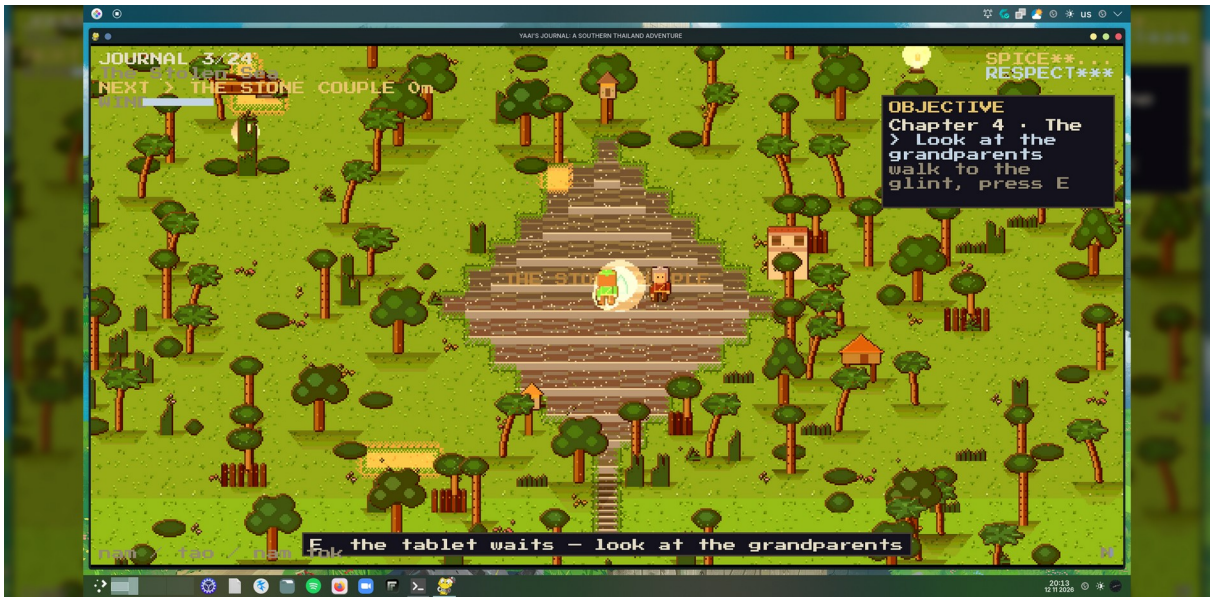
RESULTS AND DISCUSSION

3.1 Results of Yaai's Journal game development

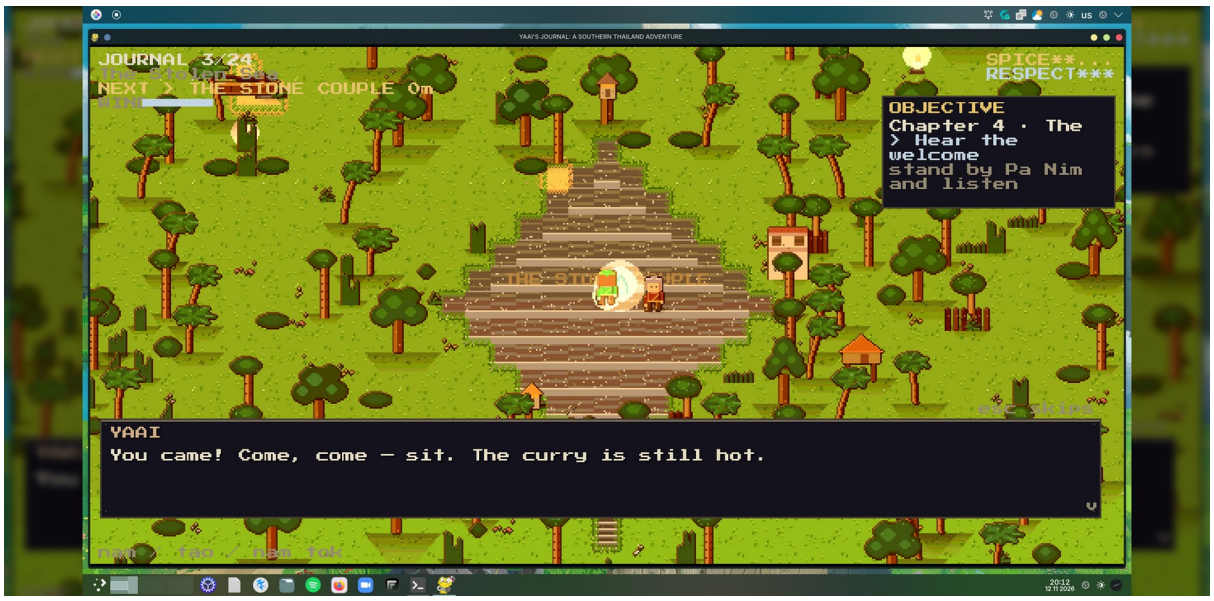
Five acts and 24 stops carry the player from Chumphon to Narathiwat. Every stop ends at a rune tablet that runs the player's own Python and refuses the pre-filled code, so the next page is earned, not clicked past. Between stops the road holds 40 heritage landmarks - the Chaiya chedi, the Phra Mahathat stupa, Phuket's shophouses, Pattani's mosque - together with 25 side quests and 74 encounters, and two cutscenes open and close the journey.



Picture 2 Cover screen - live capture of the running game (the desktop panel is visible at the bottom)



Picture 3 Overworld and quest beacon - live capture of the running game



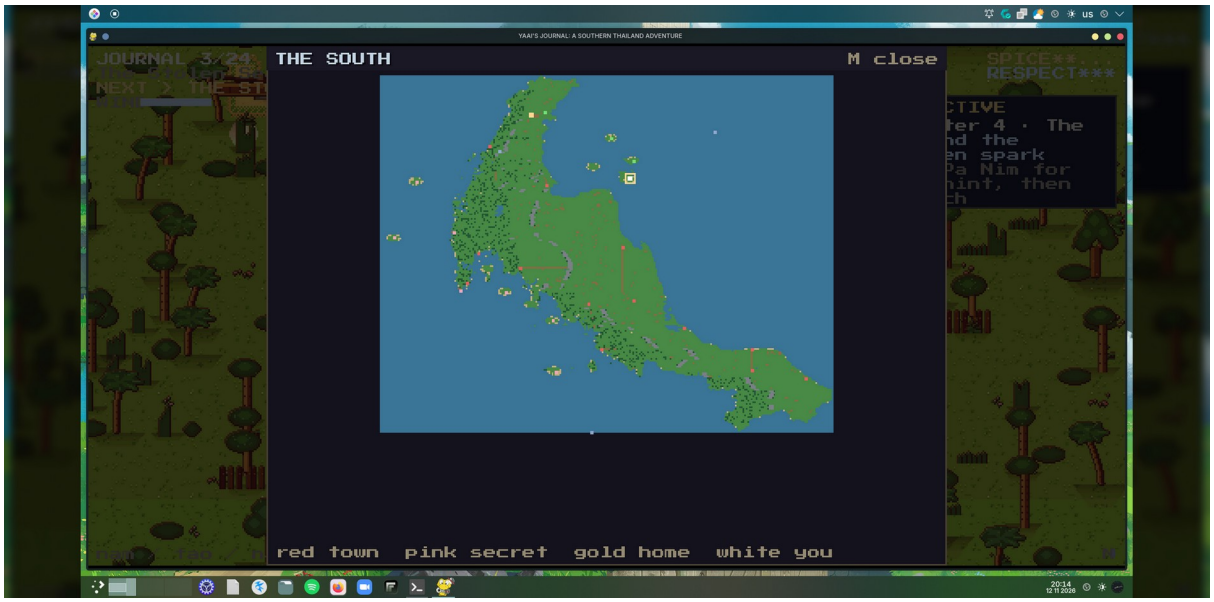
Picture 4 Dialogue with Noi - live capture of the running game



Picture 5 Rune tablet (Python puzzle) - live capture of the running game



Picture 6 The journal of collected pages - live capture of the running game



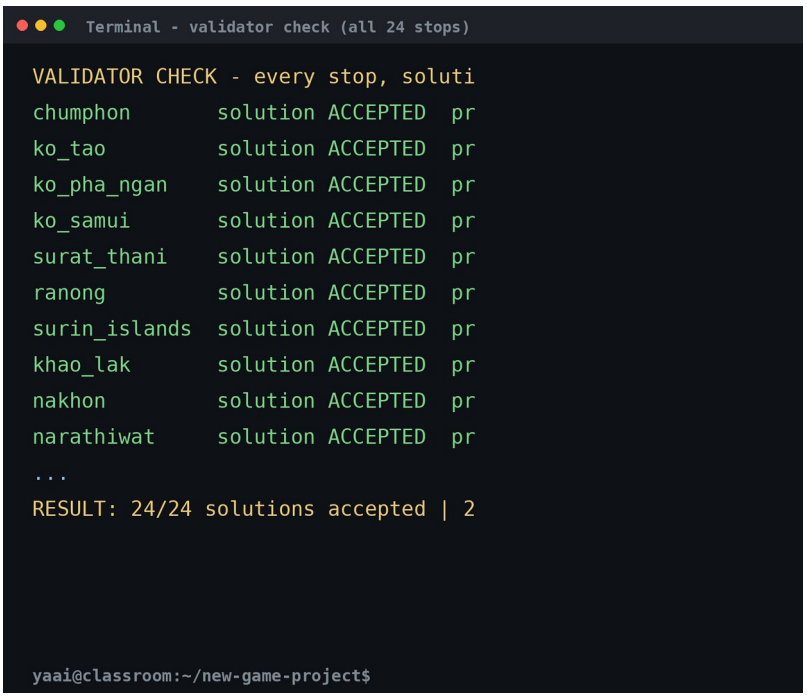
Picture 7 Sailing map - live capture of the running game

```
$ python -m unittest discover -s tests -v

test_every_track_file_exists (test_audio.TestAudio.test_every_track_file_exists) ... ok
test_mood_switching (test_audio.TestAudio.test_mood_switching) ... ok
test_music_mood_keys (test_audio.TestAudio.test_music_mood_keys) ... ok
test_scene_mapping_covers_scenes (test_audio.TestAudio.test_scene_mapping_covers_scenes) ... ok
test_base_scene_dialogue_exits (test_click.TestClick.test_base_scene_dialogue_exits) ... ok
test_canvas_pos_maps_letterbox_and_scale (test_click.TestClick.test_canvas_pos_maps_letterbox_and_scale)
A scaled window with black bars clicks the right canvas pixel. ... ok
test_center_click_remaps_at_any_window_size (test_click.TestClick.test_center_click_remaps_at_any_window_size)
A window-space click still reaches the scene at every size. ... ok
test_cover_buttons_click (test_click.TestClick.test_cover_buttons_click) ... ok
test_dialogue_exits (test_click.TestClick.test_dialogue_exits)
Esc (or right-click) bails out of any conversation. ... ok
test_finale_dialogue_exits (test_click.TestClick.test_finale_dialogue_exits) ... ok
test_host_click (test_click.TestClick.test_host_click) ... ok
test_options_rows_click (test_click.TestClick.test_options_rows_click) ... ok
   ...   (186 more cases)   ...

----------------------------------------------------------------------
Ran 201 tests in 30.521s
OK
```

Picture 8 Automated test suite - fresh run of the delivered build (201 cases, OK)



Picture 9 Validator check - 24/24 solutions accepted, 24/24 prefills rejected

Board 3

RESULTS AND DISCUSSION

3.2 Performance test results

3.2.1 Speed and stability

- 60 fps on the development laptop: the 640 x 320 canvas is scaled 4x, and one frame update plus draw costs 8.5 ms on a software rasteriser (118 fps ceiling).

- 201 test cases pass in 30.5 s, headless; two pip packages (pygame-ce and Pillow); the game runs fully offline.

- One source tree runs on Windows, Linux and macOS; the same installer is tested on every change.

3.2.2 Accuracy of the answer validator

- All 24 prepared solutions are accepted; all 24 pre-filled starter codes are rejected, so no lesson can be skipped.

3.2.3 Size and maintainability (delivered source):

- 17,762 lines in 52 game modules (18,113 with the 351-line entry point) plus 3,560 test lines in 26 test files: 22,258 lines of Python in 83 files, with 585 lines of build and capture tooling.

- Deliverables: a 29-page A4 report in English (TH SarabunPSK 16 pt), a 25-page Thai report, and this three-panel board; every picture here is a live capture of the delivered build.

- **All art is code (artgen.py, tiles.py): no image files ship.**

- **New content is one dictionary; the tests catch broken data.**

3.2.4 Utilization and Importance

- Students: a reason to code beyond a grade.
- Teachers: two packages, offline, no student data.
- School: 24 words, 40 landmarks, 25 stories to extend.

RECOMMENDATION

1. Content packs for other provinces and subjects.

2. A printable session summary for teachers.

3. A Thai/English language toggle and a CI runner for future content packs.

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QR code for the project report (Google Drive)